

WORLDSONUS: BRINGING SOUND TO WORLDS

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ABSTRACT

Recent advances in world models have enabled increasingly realistic visual synthesis. However, these generated environments remain largely silent. Bringing sound to world models poses three core challenges: **real-time generation** to keep pace with interactive video streams, **interactive control** to respond to mid-stream sound instructions, and **spatially aligned stereo** to reflect scene geometry and camera motion. To address these demands, we introduce WorldSonus, an interactive video-to-audio framework designed for real-time spatial sound synthesis in world models. For real-time generation, WorldSonus employs a streaming causal autoregressive diffusion architecture that synthesizes audio chunks at a low real-time factor (RTF) of 0.41. For interactive control, we incorporate an audio-centric captioning pipeline with chunk-indexed prompt scheduling, enabling dynamic manipulation of sound events during generation. For spatial alignment, we leverage high-quality stereo supervision curated from diverse stereo and ambisonic data. Extensive experiments demonstrate that while tailored for world models, WorldSonus generalizes effectively to open-domain video-to-audio benchmarks, matching or outperforming state-of-the-art bidirectional models in both acoustic quality and spatial alignment. Project page: <https://noizai.github.io/WorldSonus/>.

1 INTRODUCTION

Recent advances in video generation (Wang et al., 2025; Cai et al., 2025; HaCohen et al., 2025; Kong et al., 2024; Yang et al., 2025; Team, 2025b) have accelerated the development of generative world models (Valevski et al., 2025; He et al., 2025; Sun et al., 2025b; Gao et al., 2026). While these models dynamically synthesize visual environments in response to user actions, the resulting virtual worlds remain predominantly silent. Bringing realistic sound to interactive world models requires audio that seamlessly tracks an evolving visual stream, maintains camera-aligned spatial balance under dynamic viewpoints, and immediately responds to updated text instructions during an ongoing session.

Existing approaches to world-model sound generation fall into two primary categories. The first direction builds upon joint audiovisual foundation models (HaCohen et al., 2026) extended to long-horizon streaming (Duan et al., 2026; Zhang et al., 2026a; Su et al., 2026). However, coupling audio synthesis directly to the visual backbone prevents these systems from serving as standalone audio modules for visual-only world models, while causal distillation or long autoregressive (AR) rollouts often accumulate error over time. The second direction employs dedicated video-to-audio (V2A) generators, allowing modular integration with external visual engines. Yet current streaming V2A models remain constrained: V-AURA (Viertola et al., 2025) uses bidirectional visual windows and outputs mono audio. SoundReactor (Saito et al., 2025) achieves real-time stereo generation but lacks mid-stream text controllability. SwanSphere (Lei et al., 2026) focuses on panoramic video and first-order ambisonics (FOA) soundfields rather than standard perspective streams. Consequently, there remains a need for a modular V2A framework that simultaneously delivers causal real-time generation, mid-stream interactive text control, and spatially aligned perspective stereo audio.

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We present WorldSonus, a modular V2A framework designed to endow generative world models with real-time, controllable stereo audio. For **real-time generation**, WorldSonus employs a streaming AR-diffusion architecture emitting 100 ms audio chunks conditioned only on past and current visual context. A bounded persistent state ensures constant memory usage and compute cost as the stream expands. For **interactive control**, training on time-varying prompt schedules combined with chunk-level cross-attention enables users to dynamically revise text instructions for on-screen or off-screen sounds without session resets or past-frame recomputation. For **spatially aligned stereo**, we use clear-stereo clips and convert panoramic FOA data into camera-aligned stereo supervision.

Generating precise audiovisual alignment within causal contexts introduces unique challenges, as future visual frames are unavailable. To enforce temporal synchronization under limited lookahead, we introduce two-timescale visual conditioning and ShiftNCE. Chunk-level visual summaries guide the AR backbone to maintain semantic continuity, while frame-aligned local tokens condition the flow head with fine-grained spatial and temporal cues. To refine event timing without inference cost, ShiftNCE trains the AR state using a contrastive objective supervised by a frozen synchronization expert, eliminating ambiguous temporal negatives during training.

Experiments demonstrate that despite causal operation, WorldSonus matches or exceeds state-of-the-art offline bidirectional V2A baselines across both open-domain VGGSound (Chen et al., 2020) videos and our interactive benchmark, which comprises gameplay and real-world footage. WorldSonus achieves VGGish FAD (Hershey et al., 2017; Kilgour et al., 2019) scores of 1.73 on 5 s clear-stereo VGGSound (Chen et al., 2020) clips and 2.03 on 30 s interactive videos, with an execution latency of 41.2 ms per 100 ms audio chunk (real-time factor, $RTF = 0.41$) on a single NVIDIA H100 GPU.

Our main contributions are threefold:

- **Real-time, interactive stereo generation.** We introduce WorldSonus, a modular V2A framework that unifies real-time causal generation, mid-stream prompt responsiveness, and camera-aligned stereo synthesis for interactive world models.
- **Causal audiovisual synchronization.** We propose two-timescale visual conditioning and a training-only ShiftNCE distillation objective to maintain temporal synchronization under causal streaming without adding runtime latency.
- **State-of-the-art performance.** Extensive evaluations show that WorldSonus achieves competitive or superior acoustic quality and stereo balance compared with offline bidirectional baselines, with counterfactual tests confirming robust interactive control.

2 RELATED WORK

Bidirectional video-to-audio generation. Bidirectional video-to-audio (V2A) synthesis has evolved from text-to-audio latent diffusion models (Liu et al., 2023a) and early V2A diffusion networks (Luo et al., 2023; Zhang et al., 2026b) to time-aligned models and rectified flow architectures (Wang et al., 2024; Cheng et al., 2025), improving semantic and temporal fidelity through audio-visual pretraining and explicit synchronization features. Recent models such as AudioX (Tian et al., 2025), ThinkSound (Liu et al., 2025a), and PrismAudio (Liu et al., 2025b) render stereo via flow matching and reward alignment. AC-Foley (Fang et al., 2026) adds reference-audio conditioning, while Omni2Sound (Dai et al., 2026) investigates video-text-to-audio diffusion across modality combinations. Meanwhile, spatial audio generation has advanced from diffusion-based binaural synthesis (Leng et al., 2022) toward object-aware stereo and ambisonics (Karchkhadze et al., 2025; Kim et al., 2025; Liu et al., 2025c). However, these approaches require full-clip visual context and do not support streaming video input or dynamic control within a persistent causal session.

Streaming video-to-audio generation. V-AURA (Viertola et al., 2025) uses AR audio-token prediction with chunked waveform decoding, while SwanSphere (Lei et al., 2026) extends causal generation to spatial audio conditioned on panoramic video and text. Neither establishes fully end-to-end causal streaming: V-AURA conditions each audio step on a bidirectional visual window and non-causal DAC (Kumar et al., 2023) decoding, and SwanSphere’s variational autoencoder (VAE) (Kingma & Welling, 2014), based on Stable Audio Open (Evans et al., 2025), does not support stateful stream decoding. SoundReactor (Saito et al., 2025) combines a causal AR backbone with a diffusion head for streaming audio from gameplay video. Its V2A generator weights are not public,

so we adopt only the released causal VAE. Building on this line of work, we target open-domain, interactive video with bounded persistent state and sound instructions that can change mid-stream.

Interactive world models and joint audio–video generation. Interactive video world models such as GameGen-X (Che et al., 2025) and LingBot-World 2.0 (Gao et al., 2026) support controllable visual rollouts and increasingly persistent interaction. Complementary approaches, including Omni-Forcing (Su et al., 2026) and Ripple (Ding et al., 2026), jointly generate streaming audio and video. We instead treat audio as a separate causal module conditioned on an externally produced visual stream: the upstream model exposes only realized frames, without requiring access to its action space or internal architecture, while users or agents independently update audio prompts.

3 WorldSonus

WorldSonus synthesizes real-time stereo audio from streaming video and dynamic prompts via chunk-wise causal AR generation (Figure 1). It couples a bounded recurrent memory with multi-scale visual conditioning and contrastive temporal alignment.

3.1 STREAMING FORMULATION

At each streaming chunk $t \in \{1, \dots, T\}$, the model receives visual frames $\mathbf{v}_t$ and active prompt $\mathbf{p}_t$ to predict the corresponding audio latent chunk $\mathbf{a}_t$. Conditioned on a bounded history state $\mathbf{M}_{t-1}$, the joint distribution factorizes causally over time:

$$p_\theta(\mathbf{a}_1, \dots, \mathbf{a}_T) = \prod_{t=1}^T p_\theta(\mathbf{a}_t \mid \mathbf{M}_{t-1}, \mathbf{v}_t, \mathbf{p}_t), \quad (1)$$

where $\mathbf{M}_{t-1}$ is maintained by a sliding-window Key-Value cache bounded by maximum length W . This factorization uses only frames of past and current chunks, without access to frames of future blocks. Audio is synthesized in the continuous latent space of a frozen causal stereo VAE (Saito et al., 2025), which encodes 48 kHz stereo audio into latents advancing at 30 Hz. Its stateful decoder converts predicted chunks to waveform during streaming, with no re-decoding of past latents. Each 100 ms generation chunk corresponds to three latent frames $\mathbf{a}_t \in \mathbb{R}^{3 \times d_a}$, time-aligned with the three visual frames in $\mathbf{v}_t$ (30 FPS).

3.2 AUTOREGRESSIVE DIFFUSION

To balance long-range semantic continuity with high-fidelity continuous generation, we adopt a chunked AR-diffusion framework (Li et al., 2024; Saito et al., 2025; Lei et al., 2026). At each step, a decoder-only Transformer processes two chunk-aggregated tokens (one from the previous audio chunk and one from the current video chunk) to maintain session state. Trained with a causal sliding-window attention mask, the model deploys a fixed-capacity Ring-KV cache over context window W (a circular buffer overwriting the oldest slots in place). This ensures bounded memory while matching the context-window policy between training and inference.

The resulting AR hidden state $\mathbf{h}_t$ conditions a compact flow head, which denoises the three latent frames of chunk t using a rectified-flow objective (Lipman et al., 2023; Liu et al., 2023b). The flow head attends bidirectionally inside the current chunk, without access to frames of future blocks, and it reads the frame-aligned visual tokens of Section 3.3 as finer-grained spatial and temporal evidence. Appendix A gives the remaining network details.

3.3 TWO-TIMESCALE VISUAL CONDITIONING

To reconcile long-horizon semantic continuity in the bounded AR backbone with precise acoustic rendering in the flow head, we decouple visual conditioning into two complementary timescales: **chunk-level semantics** and **frame-aligned local features**. For each chunk (spanning three frames), a frozen DINOv3 (Siméoni et al., 2026) encoder maps video frames into spatial patch grids $\mathbf{G}_n$. Mixed-aspect inputs are assigned to resolution buckets and resized without stretching at a fixed pixel budget. To capture motion cues, each grid is concatenated with its temporal difference $\Delta \mathbf{G}_n = \mathbf{G}_n - \mathbf{G}_{n-1}$ (Chen et al., 2026). A learned query then aggregates each frame’s spatial grid into a single feature token, yielding three frame tokens per chunk.

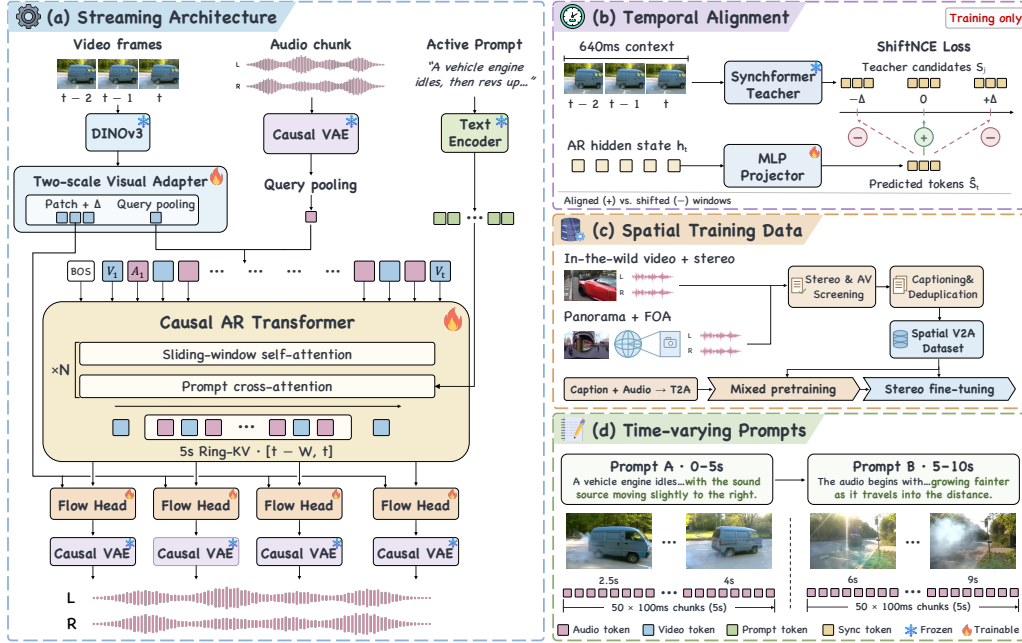


Figure 1: Overview of WorldSonus. (a) **Causal streaming pipeline**: Video frames map to two-timescale visual tokens conditioning an AR transformer and a rectified-flow head. (b) **Training-only ShiftNCE**: A training-only frozen Synchformer teacher guides temporal alignment via contrastive window matching. (c) **Spatial stereo supervision**: In-the-wild stereo filtering and panoramic FOA view-decoding provide directional audio grounding. (d) **Interactive prompt control**: Text prompts update dynamically at chunk boundaries, preserving session state and acoustic continuity.

These frame tokens are then processed into the two target timescales: For **chunk-level semantics**, a learned summary query aggregates the three frame tokens into a single chunk-level visual token, which is routed to the AR backbone to maintain a compact footprint. For **frame-aligned local features**, the three refined frame tokens bypass the AR backbone to condition the flow head directly. Our dual-path architecture provides the flow head with frame-aligned visual tokens without further chunk-level aggregation, which is essential for preserving the temporal granularity of individual video frames.

3.4 INTERACTIVE PROMPT CONTROL

To support interactive control during streaming, we introduce an in-place prompt control mechanism that allows text instructions to be updated mid-stream.

Given a sound instruction, a T5Gemma 2 (Zhang et al., 2025) text encoder and compressor project it into a compact token set that conditions the AR backbone via cross-attention, which is cached across streaming steps. When an instruction updates during streaming, the system performs an in-place replacement of the cross-attention cache at the nearest chunk boundary, preserving the Ring-KV cache and existing visual and audio states. To train the model for these dynamic transitions, we incorporate time-varying prompt schedules including mid-sequence instruction switches and drops. This enables seamless steering in a persistent causal session.

3.5 TRAINING

Teacher-forced pretraining. The generator is pretrained under a rectified flow objective on a mixture of stereo video-audio and audio-only data, using teacher forcing over the bounded context window W . For a ground-truth latent chunk $\mathbf{a}_t$, noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and flow timestep $s \sim \mathcal{U}[0, 1]$, the linear interpolation state is $\mathbf{x}_{t,s} = (1 - s)\epsilon + s\mathbf{a}_t$. The velocity matching loss is formulated as:

$$\mathcal{L}_{\text{flow}}(\epsilon) = \|f_{\theta}(\mathbf{x}_{t,s}, s \mid \mathbf{C}_t) - (\mathbf{a}_t - \epsilon)\|_2^2, \quad (2)$$

where $\mathbf{C}_t$ aggregates the AR history state and local frame-aligned visual slots. We apply Explorative Modeling (Gladstone et al., 2026) to flow-matching training by evaluating three candidate source noises per step and optimizing the best-matching draw.

Teacher-guided temporal alignment. While existing synchronization models (Iashin et al., 2024) provide strong temporal cues, deploying them at inference introduces extra latency and windowing constraints. We instead employ a frozen synchronization encoder as a training-only teacher to distill temporal alignment directly into the generator.

Let $\mathbf{S}_t$ denote the teacher’s target embedding for a video window ending at chunk t , and let a projector $P(\cdot)$ map the generator’s internal AR state $\mathbf{h}_t$ to predict $\mathbf{S}_t$. To enforce temporal synchronization, our ShiftNCE objective contrasts the aligned target $\mathbf{S}_t$ against temporally shifted embeddings $\mathbf{S}_{t+\delta}$ from the same video across a set of non-zero chunk offsets $\delta \in \mathcal{D}_t$:

$$\mathcal{L}_{\text{sync}} = -\log \frac{\exp(\text{sim}(P(\mathbf{h}_t), \mathbf{S}_t)/\tau)}{\exp(\text{sim}(P(\mathbf{h}_t), \mathbf{S}_t)/\tau) + \sum_{\delta \in \mathcal{D}_t} \exp(\text{sim}(P(\mathbf{h}_t), \mathbf{S}_{t+\delta})/\tau)}, \quad (3)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, and τ is the temperature hyperparameter. To prevent false negatives during static visual segments, temporally shifted candidates with high visual similarity under the teacher model are filtered out. The final pretraining objective is the joint loss $\mathcal{L} = \mathcal{L}_{\text{flow}} + \lambda_{\text{sync}} \mathcal{L}_{\text{sync}}$, where λ_{sync} balances the two objectives. Following pretraining, we fine-tune the network on a high-quality stereo video-audio subset. Complete training hyperparameter settings are detailed in Appendix B.

4 DATA CURATION

Data sources. Our dataset contains 1,465 hours of audio, including 999 hours of paired stereo video-audio data and 466 hours of audio-only data. The *video-audio subset* uses open-source datasets including VGGSound (Chen et al., 2020), AudioSet (Gemmeke et al., 2017), Kinetics-700 (Carreira et al., 2019), and HD-EPIC (Perrett et al., 2025) to cover a wide range of acoustic events. To support audio generation for interactive world models, we collect additional standard stereo video with clear camera motion and dynamic objects. We also incorporate panoramic recordings from the existing Sphere360 (Liu et al., 2025c) and YT-AmbiGen (Kim et al., 2025) datasets. For panoramic clips, we estimate horizontal sound directions from the acoustic energy field to sample perspective visual crops and render view-aligned stereo audio. The *audio-only subset* includes AudioCaps (Kim et al., 2019) and isolated audio tracks from weakly-aligned video clips, where the visual content is not aligned with the audio, but the soundtrack retains high-quality acoustic events.

Data filtering. To ensure spatial quality and video-audio alignment, we use a two-stage filtering pipeline. First, automated signal processing removes silent, phase-corrupted, narrow-stereo, and dual-mono samples. Second, Qwen3-Omni (Team, 2025a) acts as a verifier to filter out severe cross-modal mismatches and non-diegetic audio like voiceovers and background music. Complete details appear in Appendix C.

Captioning. We generate dense audio-centric captions using Qwen3-Omni (Team, 2025a). Prompts focus on sound categories, acoustic texture, temporal timing, and spatial ambiance, while excluding visual details unrelated to sound to avoid introducing irrelevant conditioning cues. Captions are generated from both video and audio for video-audio data, and from audio alone for audio-only data. Annotation details are in Appendix C.3.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Evaluation sets. We evaluate mainly on five test splits: two clear-stereo VGGSound (Chen et al., 2020) splits (5 s and 10 s; 4,096 clips each) and three splits comprising interactive gameplay and real-world footage with dynamic camera motion (5 s and 10 s, with 4,096 clips each; 30 s, with 1,024 clips). We additionally evaluate on physical collision events from Greatest Hits (Owens et al., 2016).

Table 1: Quantitative comparison across five held-out splits (4,096 clips for 5 s/10 s; 1,024 for 30 s). Best results are in bold, and second-best are underlined. Acc.: Bi (Bidirectional), Str (Stream), Caus (Causal). Cond.: VT (video+text), V (video-only). Chunk/*t*: chunk length and compute time in ms. IB and CLAP are multiplied by 100. S-FD_O is omitted for monophonic V-AURA.

Set	Model	Acc.	Cond.	Chunk/ <i>t</i>	Distribution (mid)				Spatial	Semantic		Temporal
					FAD↓	FD _P ↓	FD _O ↓	KL _P ↓	S-FD _O ↓	IB↑	CLAP↑	DeSync↓
VGG 5 s	AudioX	Bi	VT	–	3.00	153.29	38.19	<u>1.66</u>	87.60	<u>26.96</u>	37.87	0.986
	ThinkSound	Bi	VT	–	2.94	<u>141.40</u>	47.32	1.91	<u>62.95</u>	24.12	33.62	0.481
	PrismAudio	Bi	VT	–	<u>2.09</u>	132.58	50.72	1.67	94.77	26.28	39.20	<u>0.539</u>
	V-AURA	Str	V	640/636	4.06	304.06	51.01	2.03	–	26.48	24.91	1.287
	WorldSonus (Ours)	Caus	VT	100/41.2	1.73	159.74	<u>39.03</u>	1.48	43.47	27.06	<u>38.35</u>	0.686
VGG 10 s	AudioX	Bi	VT	–	3.86	194.72	41.50	1.54	83.53	27.84	<u>34.56</u>	0.944
	ThinkSound	Bi	VT	–	<u>2.18</u>	144.17	46.49	1.73	<u>55.02</u>	25.76	30.46	0.514
	PrismAudio	Bi	VT	–	2.64	<u>154.69</u>	52.38	<u>1.47</u>	118.08	<u>28.16</u>	33.26	0.427
	V-AURA	Str	V	640/636	4.35	318.13	57.71	1.95	–	26.73	23.35	1.208
	WorldSonus (Ours)	Caus	VT	100/41.2	1.79	167.52	<u>41.90</u>	1.32	45.40	28.82	36.77	0.687
Inter. 5 s	AudioX	Bi	VT	–	4.44	<u>186.49</u>	66.77	<u>1.56</u>	109.21	24.88	<u>32.53</u>	1.059
	ThinkSound	Bi	VT	–	8.49	256.38	<u>61.84</u>	1.98	<u>58.68</u>	18.39	22.15	0.733
	PrismAudio	Bi	VT	–	6.22	222.83	<u>62.74</u>	1.60	83.85	23.08	32.42	0.651
	V-AURA	Str	V	640/636	8.18	436.10	83.78	2.01	–	26.84	16.07	1.344
	WorldSonus (Ours)	Caus	VT	100/41.2	2.68	163.74	40.99	1.29	34.26	<u>26.48</u>	35.90	0.831
Inter. 10 s	AudioX	Bi	VT	–	<u>5.05</u>	352.02	69.19	1.57	119.16	25.39	27.89	1.077
	ThinkSound	Bi	VT	–	6.27	<u>332.49</u>	<u>60.93</u>	1.85	<u>56.47</u>	21.43	24.10	0.738
	PrismAudio	Bi	VT	–	5.16	<u>338.17</u>	<u>61.23</u>	<u>1.42</u>	96.12	25.20	<u>31.08</u>	0.599
	V-AURA	Str	V	640/636	9.75	532.74	86.46	1.96	–	27.60	15.47	1.192
	WorldSonus (Ours)	Caus	VT	100/41.2	2.62	262.25	42.91	1.30	37.42	28.63	34.22	0.820
Inter. 30 s	AudioX	Bi	VT	–	<u>5.52</u>	427.00	81.14	1.88	121.62	24.78	30.34	1.125
	ThinkSound	Bi	VT	–	7.00	473.40	73.46	2.31	<u>81.88</u>	11.71	19.50	0.901
	PrismAudio	Bi	VT	–	6.08	<u>375.22</u>	<u>62.99</u>	<u>1.66</u>	95.26	22.32	<u>33.70</u>	0.765
	V-AURA	Str	V	640/636	13.33	666.31	108.01	2.74	–	19.93	8.54	1.266
	WorldSonus (Ours)	Caus	VT	100/41.2	2.03	253.21	26.67	1.22	25.44	<u>22.93</u>	34.21	<u>0.867</u>

Baselines. WorldSonus is a causal model, using only frames from past and current chunks without access to future frames. In the absence of causal streaming stereo baselines, we evaluate against bidirectional stereo models including AudioX (Tian et al., 2025), ThinkSound (Liu et al., 2025a), and PrismAudio (Liu et al., 2025b), alongside the streaming monophonic baseline V-AURA (Viertola et al., 2025).

5.2 METRICS

Audio quality and audiovisual alignment. For *acoustic fidelity*, we report Fréchet distances in VGGish (Hershey et al., 2017) (FAD (Kilgour et al., 2019)), PaSST (Koutini et al., 2022) (FD_P), and OpenL3 (Cramer et al., 2019) (FD_O) feature spaces, together with PaSST-based KL divergence (KL_P). FAD and FD_O use the mid-channel signal. S-FD_O applies the OpenL3 distance to the side-channel signal to evaluate stereo distribution quality. For *audiovisual alignment*, we use ImageBind (Girdhar et al., 2023) (IB) similarity for semantic consistency and Synchformer (Iashin et al., 2024) offset (DeSync) for synchronization, supplemented by onset accuracy, F1, and AP on Greatest Hits following MMAudio (Cheng et al., 2025).

Stereo balance agreement. BiasSkill measures whether generated and reference audio favor the same left/right channel within 1 s windows. It reports the normalized gain over a permutation baseline that accounts for each model’s fixed channel bias. Details are provided in Appendix E.

5.3 MAIN RESULTS

Audio quality. We conduct quantitative comparisons across all five held-out test splits. As shown in Table 1, WorldSonus achieves competitive or superior acoustic quality (FAD, FD_P, and KL_P) compared to bidirectional models on both VGGSound and interactive video datasets, despite using only visual frames from past and current chunks. Compared with the streaming baseline V-AURA, WorldSonus achieves lower distributional divergence and stronger audiovisual alignment while

running at a much finer temporal resolution (100 ms vs. 640 ms), synthesized in just 41.2 ms per chunk on a single NVIDIA H100 GPU (RTF = 0.41) to comfortably satisfy real-time streaming requirements. Together, these results demonstrate that bounded-memory causal generation can combine competitive audio quality with real-time synthesis across the evaluated 5–30 s durations.

Spatial and temporal alignment. For spatial alignment, we complement distributional stereo metrics (S-FD_O in Table 1) with stereo balance agreement measured by BiasSkill over 1 s windows (Table 2). WorldSonus achieves the highest BiasSkill across all evaluated splits. Figure 2 provides complementary qualitative evidence from two interactive clips with sound sources located predominantly on one side: beach surf on the left and a passing truck on the right. The left/right spectrograms and signed energy-balance curves ($(E_R - E_L)/(E_R + E_L)$) show that WorldSonus reproduces the corresponding channel dominance, consistent with the ground-truth audio, whereas the baselines exhibit weaker or less consistent spatial alignment.

Table 2: Stereo balance agreement (BiasSkill, %). Full table in App. E.

Model	VGG 10 s	Inter. 10 s	Inter. 30 s
AudioX	3.79	3.08	−0.01
ThinkSound	3.22	−1.78	0.63
PrismAudio	1.10	−0.06	4.19
WorldSonus (Ours)	4.42	6.03	15.90

Table 3: Onset detection on Greatest Hits.

Model	Acc.↑	F1↑	AP↑
AudioX	0.858	0.770	0.790
ThinkSound	0.706	0.758	0.904
PrismAudio	0.783	0.808	0.906
w/o finetune	0.805	0.796	0.884
w/o ShiftNCE	0.678	0.560	0.681
WorldSonus (Ours)	0.803	0.785	0.871

For temporal alignment, bidirectional baselines that optimize directly on Synchformer features achieve lower DeSync (Table 1). Nevertheless, WorldSonus maintains competitive event timing without test-time guidance, as reflected in Table 3 on the model-free Greatest Hits onset benchmark.

Together, these results show that WorldSonus combines competitive temporal alignment with stronger spatial alignment under causal visual conditioning.

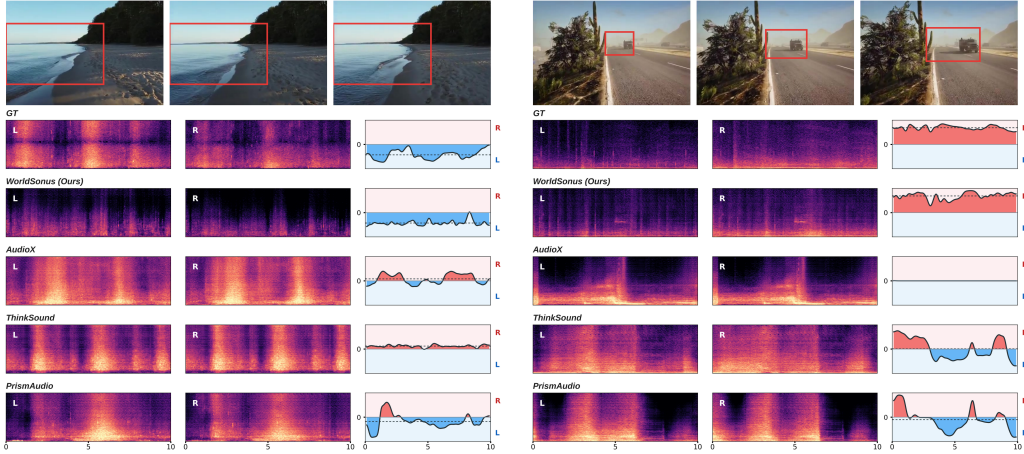


Figure 2: Qualitative stereo layout versus bidirectional baselines on two interactive clips. Red boxes mark sources. Energy-balance curves show left (negative) versus right (positive) channel dominance.

Long-horizon stability. To analyze potential drift during continuous streaming, we evaluate 1,024 Interactive 30 s clips under two regimes: *Direct* (generating the final 25–30 s window from a cold start) and *Rollout tail* (extracting the final window from an uninterrupted 30 s stream). As shown in Table 4, the rollout tail closely tracks the direct baseline in acoustic fidelity and synchronization (FAD 2.51 vs. 2.63, DeSync 0.827 vs. 0.880), with minimal shifts in OpenL3 and ImageBind. It indicates that under our bounded 5 s Ring-KV cache, WorldSonus maintains stable, collapse-free streaming across extended horizons without periodic resets.

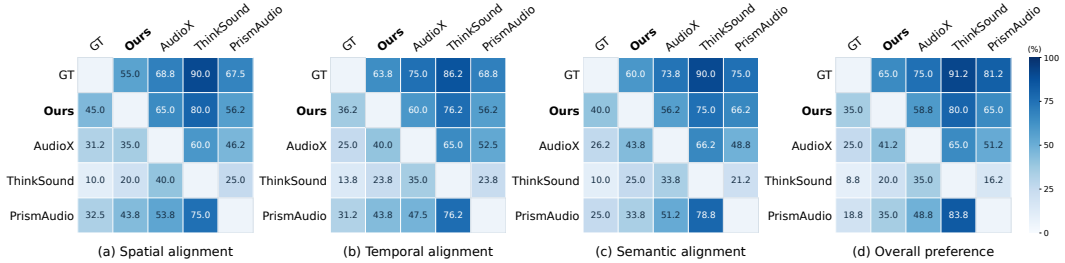
Table 4: Long-horizon stability on Interactive 30 s (1,024 clips). Final 5 s window (25–30 s) generated directly versus via continuous rollout.

Protocol	FAD↓	FD _P ↓	FD _O ↓	KL _P ↓	S-FD _O ↓	IB↑	DeSync↓
Direct last 5 s	2.63	271.68	35.16	1.22	39.24	21.38	0.880
Rollout tail (25–30 s)	2.51	246.69	40.46	1.31	41.77	19.99	0.827

Text control. To evaluate interactive prompt control, we measure text to audio agreement using CLAP (Wu et al., 2023) between each 5 s audio segment and its caption. For 10 s clips, let P_1 and P_2 denote the sequential instructions for the first and second 5 s halves ($P_1 \rightarrow P_2$). We report the dual relative match rate, defined as the percentage of clips where both 5 s segments score higher with their own caption than with the other. As shown in Table 5, WorldSonus achieves dual match rates of 25.68% on VGGSound and 23.44% on Interactive 10 s, outperforming all baseline architectures.

However, strong text-audio alignment alone does not fully isolate text responsiveness from natural visual transitions. To rule out visual conflation, we perform paired counterfactual interventions by comparing a Switch setting ($P_1 \rightarrow P_2$ at 5 s) against a Hold baseline ($P_1 \rightarrow P_1$) under identical visual features and seeds. The intervention gain G measures how much switching increases the second half’s CLAP preference for P_2 over P_1 . As shown in Table 6, WorldSonus yields positive net intervention gains G (+0.0509 and +0.0525) with 95% bootstrap confidence intervals excluding zero. These combined results confirm that WorldSonus achieves active, causal text steering over generated audio with visual context held fixed. Details are provided in Appendix F.

Subjective comparison. To complement objective evaluations, we perform a blind, randomized human A/B study on 40 representative Interactive 10 s clips. Twenty assessors provided 400 pairwise evaluations across four criteria: spatial alignment, temporal alignment, semantic match, and overall preference, with ties counted as half a vote (Appendix G). As shown in Figure 3, WorldSonus is preferred over all three generated baselines across all criteria, while ground-truth audio remains preferred. Overall preference reaches 58.8% against AudioX, 80.0% against ThinkSound, and 65.0% against PrismAudio, indicating stronger overall listener preference.

Figure 3: User-study preference matrices across (a) spatial alignment, (b) temporal alignment, (c) semantic alignment, and (d) overall preference. Cell (i, j) is the percentage preferring the row method over the column method, counting each tie as half a vote (40 ratings per pair).

5.4 ABLATION STUDIES

All ablation checkpoints are evaluated directly after pretraining without fine-tuning on the Interactive 10 s set (4,096 clips) under fixed classifier-free guidance (Ho & Salimans, 2022). Table 7 isolates architectural and alignment components, while Table 8 compares generation-chunk granularities.

Temporal alignment objective. To investigate whether the proposed ShiftNCE objective explicitly drives fine-grained synchronization, and whether external temporal representations can replace it, we compare our default model against a variant trained without ShiftNCE and another incorporating direct features from a 640 ms causal Synchformer (Iashin et al., 2024) window. As shown in Table 7, removing ShiftNCE degrades DeSync from 0.831 to 1.067, while direct feature injection yields worse timing and acoustic quality, possibly reflecting a mismatch between its 640 ms feature window and our 100 ms streaming chunks. These results confirm that ShiftNCE provides the necessary temporal supervision to sharpen alignment without requiring additional sync encoders or inference overhead.

Visual feature representation. To examine whether our DINOv3 patch-based representation outperforms the pooled CLIP-family features commonly used in prior V2A models (Zhang et al., 2026b; Cheng et al., 2025; Liu et al., 2025a; Shan et al., 2025), we compare it with pooled SigLIP 2 (Tschannen et al., 2025) features. We also evaluate a variant without temporal deltas (ΔG_n). Table 7 shows that our DINOv3 features substantially improve ImageBind alignment from 24.56 to 28.05 over the Pooled SigLIP 2 baseline. Furthermore, removing temporal deltas worsens both DeSync (0.958) and FAD (3.08). These results demonstrate the effectiveness of our DINOv3-based visual conditioning and explicit temporal cues for video-to-audio synthesis.

Two-timescale visual conditioning. To evaluate the role of feeding direct frame-level visual features to the flow head, we compare our model with a variant that removes this shortcut path and passes visual information only through the autoregressive state. As shown in Table 7, removing the direct visual connection leads to a severe degradation in audio quality, with FAD increasing from 2.42 to 15.82. This confirms that in our streaming design, combining direct frame-level visual inputs with higher-level autoregressive context is essential for generating clear and synchronized audio.

Generation-chunk granularity. To determine the optimal balance between streaming latency and generation quality, we evaluate chunk sizes across 33 ms, 100 ms, and 200 ms. Table 8 shows that shrinking the chunk size to 33 ms worsens acoustic quality (FAD 3.79), whereas increasing it to 200 ms increases latency without consistent improvements across metrics. We therefore choose 100 ms as a practical balance between streaming granularity and generation quality.

6 CONCLUSION

We present WorldSonus, a causal spatial audio synthesis framework for real-time sound generation in interactive world models. By factorizing generation into chunk-wise autoregressive flow matching, WorldSonus synthesizes 48 kHz stereo audio in 100 ms chunks with an execution latency of 41.2 ms per chunk on a single NVIDIA H100 GPU (RTF = 0.41). Our decoupled visual conditioning couples long-term planning via a bounded 5 s Ring-KV cache with frame-aligned local guidance in the flow head, while training-only ShiftNCE enforces precise temporal synchronization with zero inference overhead. Evaluated on open-domain videos and interactive environments, WorldSonus achieves acoustic fidelity and spatial localization competitive with or exceeding offline bidirectional models, while seamlessly supporting real-time prompt updates.

Table 7: Architecture ablations on Interactive 10 s.

Variant	FAD↓	FD _P ↓	FD _O ↓	KL _P ↓	S-FD _O ↓	IB↑	DeSync↓
w/o ShiftNCE	2.49	276.68	48.50	1.44	54.08	27.20	1.067
Direct sync features	2.81	278.28	46.33	1.40	48.68	26.22	1.043
w/o temporal delta	3.08	267.51	49.37	1.40	68.76	28.77	0.958
Pooled SigLIP 2	3.29	291.60	31.96	1.33	29.81	24.56	0.944
w/o flow-head DINO	15.82	906.49	117.12	2.25	127.59	12.65	1.020
WorldSonus (Ours)	2.42	266.41	40.97	1.31	40.17	28.05	0.831

Table 8: Chunk-length comparison on Interactive 10 s.

Chunk	FAD↓	FD _P ↓	FD _O ↓	KL _P ↓	S-FD _O ↓	IB↑	DeSync↓
33 ms	3.79	333.05	62.17	1.68	62.16	27.18	0.915
200 ms	2.51	274.86	39.03	1.26	35.12	26.35	0.819
100 ms	2.42	266.41	40.97	1.31	40.17	28.05	0.831

7 LIMITATIONS

Causal audio representations. WorldSonus generates audio in the continuous latent space of a frozen causal stereo VAE adapted from SoundReactor (Saito et al., 2025). SoundReactor (Saito et al., 2025) reports lower reconstruction quality for its causal VAE decoder than for its non-causal counterpart. Developing more expressive, large-scale causal audio codecs will be crucial to close this performance gap in future work.

Stereo data scale. High-quality spatial audio data remains scarce compared to massive monophonic corpora. Many public stereo videos contain artificial or noisy channel separation that does not accurately reflect visual object motion. While our two-stage filtering pipeline and panoramic soundfield decoding extract reliable spatial supervision, expanding the scale of authentic stereo data remains a key goal for future work.

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A IMPLEMENTATION CONFIGURATION AND ARCHITECTURAL DETAILS

Table 9 summarizes the concrete configuration and architectural choices of WorldSonus, supplementing the high-level system description in Section 3.

Table 9: System implementation and architectural configuration of WorldSonus.

Component	Configuration and Architectural Specification
Streaming Unit	100 ms chunk: 3 latent frames and 3 video frames at 30 FPS
Causal Stereo VAE	Frozen SoundReactor (Saito et al., 2025) VAE; maps 48 kHz stereo audio to 30 Hz latents
AR Context Window	Ring-KV cache over $W = 50$ chunks (5 s)
AR Backbone	36 layers, width 1,024, 16 attention heads, SwiGLU (Shazeer, 2020) hidden size 4,096
Visual Encoder	Frozen DINOv3 S+ (Siméoni et al., 2026); aspect-preserving resolution buckets; patch grid pooled $2\times$
Visual Adapter	Linear bottleneck to 128 channels, concatenated with $\Delta \mathbf{G}_n$; 2D-RoPE (Su et al., 2024; Heo et al., 2024); 1 learned query per frame plus 1 chunk-summary query
Text Prompt Encoder	Frozen T5Gemma 2 (Zhang et al., 2025), truncated at 192 tokens
Prompt Compressor	1 global token + 31 learned queries, cached until the prompt updates
Flow Head	Width 1,024, SwiGLU hidden size 5,120; bidirectional attention within chunk; cross-attention to 3 frame tokens; AdaLN-Zero (Peebles & Xie, 2023) conditioned on $\mathbf{h}_t$
Sampler	15 Euler steps per 100 ms chunk
Guidance	AR stage: $s_v = s_p = 3.0$. Flow head: AR-summary scale 1.0, local-visual scale 8.0 with APG (Sadat et al., 2025) ($\eta = 0$, momentum +0.25, radius 0.124)
ShiftNCE	Training-only objective, using a frozen Synchformer (Iashin et al., 2024) teacher

AR token pairing. At each step, the model receives the audio summary from the previous chunk and the visual summary from the current chunk. The audio summary is causally masked from attending to current or future visual tokens. Positional encoding uses chunk time indices via 2D-RoPE, maintaining continuous temporal representations when the Ring-KV buffer wraps.

Guidance formulations. Visual and textual controls are guided independently using classifier-free guidance (Ho & Salimans, 2022). Sampling evaluates a conditional branch f , a null-video branch f_{-v} , and a null-prompt branch f_{-p} :

$$\hat{f} = f + (s_v - 1)(f - f_{-v}) + (s_p - 1)(f - f_{-p}). \quad (4)$$

Guidance scales are listed in Table 9. On the flow head, APG (Sadat et al., 2025) is applied to the clean latent estimate $\hat{\mathbf{a}}_t = \mathbf{x}_{t,s} + (1 - s)\mathbf{u}$, where $\mathbf{u}$ is the predicted velocity.

Baseline evaluations and clip windows. For fair comparison, AudioX (Tian et al., 2025), ThinkSound (Liu et al., 2025a), and PrismAudio (Liu et al., 2025b) receive identical audio-centric captions as WorldSonus and are executed via their official inference pipelines with chain-of-thought reasoning disabled. V-AURA (Viertola et al., 2025) operates in video-only mode. On 5 s and 10 s benchmarks, all baseline models generate full-length clips in a single inference call. On the 30 s benchmark, AudioX and PrismAudio generate three contiguous 10 s clips matching their maximum context limits, whereas ThinkSound and V-AURA process the entire 30 s sequence in a single call. In contrast, WorldSonus continuously streams 100 ms chunks under bounded Ring-KV state with six sequential 5 s audio prompts. For 10 s caption-conditioned baseline runs, prompt pairs are concatenated in chronological order (e.g., “First five seconds: ... Next five seconds: ...”).

B TRAINING RECIPE AND OPTIMIZATION DETAILS

Training consists of two distinct stages (Table 10). Full optimization hyperparameters are listed in Table 11. The audio VAE (Saito et al., 2025), DINOv3 S+ (Siméoni et al., 2026), T5Gemma 2 (Zhang et al., 2025), and the Synchformer (Iashin et al., 2024) teacher remain frozen throughout training.

Table 10: Training curriculum: corpora, task mixtures, and view lengths.

Stage		Corpus Composition	Task Ratio (video-audio:audio-only)	View Representation
Pretraining (0–140k)		Full video-audio/audio-only mixture: base stereo datasets plus mono-dominant expansion	2:1	5 s blocks, with 10 s paired views on 4 of 5 epochs
Fine-tuning (140–150k)		High-synchrony filtered stereo video-audio only (expansion excluded)	1:0	5 s blocks, with 10 s paired views on 4 of 5 epochs

Table 11: Training configuration and hyperparameter settings.

Hyperparameter	Setting / Value
Training Steps	140,000 pretraining steps + 10,000 fine-tuning steps (150,000 total)
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.99$, weight decay 0.05)
Learning Rate	Linear warmup from 0 to 5×10^{-5} over 4,000 steps; constant 5×10^{-5} thereafter
Gradient Clipping	Maximum gradient norm 1.0
EMA Rate	Decay rate 0.9999
Distributed Setup	16 NVIDIA H100 GPUs (2 nodes); batch size 16 per GPU (256 global batch)
Flow Batch Multiplier	4 independent flow timesteps sampled per audio instance
Explorative Modeling	$K = 3$ candidate source noise vectors per example; backpropagation through best candidate
AR Causal Context	5.0 s sliding window (50 chunks) under teacher forcing
Guidance Dropout	Video dropout 0.1, prompt dropout 0.1; flow-head categorical dropout 0.2 (0.1 AR summary, 0.1 local visual); most recent teacher-forced audio token nulled with 0.2 probability
Loss Objectives	Joint Flow Matching ($\mathcal{L}_{\text{flow}}$) and Causal ShiftNCE ($\mathcal{L}_{\text{sync}}$)

B.1 EXPLORATIVE MODELING FOR FLOW MATCHING (XM3)

During pretraining, we apply Explorative Modeling (Gladstone et al., 2026) to the flow-matching objective. For each training instance, we sample $K = 3$ noises with the same timestep $s \sim \mathcal{U}[0, 1]$ and dropout mask, then compute their vector field errors without gradients:

$$\ell^{(k)} = \left\| \hat{\mathbf{u}}^{(k)} - (\mathbf{a}_t - \boldsymbol{\epsilon}^{(k)}) \right\|_2^2, \quad (5)$$

and execute backpropagation solely on $k^* = \arg \min_k \ell^{(k)}$. Standard Gaussian noise sampling is used during inference.

B.2 AMBIGUITY-AWARE CAUSAL SHIFTNCE

ShiftNCE distills synchronization signals from a frozen Synchformer (Iashin et al., 2024) teacher into the AR backbone during training, and is disabled during inference.

Teacher representations. The teacher processes a causal 640 ms visual window ending on a chunk boundary, outputting eight ordered tokens $\mathbf{S}_t$. A 2-layer MLP maps the AR representation $\mathbf{h}_t$ to this alignment space. Pairwise similarity averages cosine similarity across the eight token slots:

$$\text{sim}(P(\mathbf{h}_i), \mathbf{S}_j) = \frac{1}{8} \sum_{k=1}^8 \cos(P(\mathbf{h}_i)_k, (\mathbf{S}_j)_k). \quad (6)$$

Negative mining and anchor selection. For chunk t , candidate negatives comprise same-clip temporal offsets $\delta \in \{\pm 1, \pm 2, \pm 4\}$. Candidates exhibiting high teacher similarity ($\text{sim}(\mathbf{S}_t, \mathbf{S}_{t+\delta}) \geq 0.97$) are filtered out to avoid false-negative penalization. Among valid sequence anchors, we select the top 40% most teacher-separable anchors ranked by contrastive margin.

Objective formulation. With a fixed temperature $\tau = 0.07$, the alignment loss is averaged uniformly over selected valid anchors $\mathcal{A}_t$:

$$\mathcal{L}_{\text{sync}} = -\frac{1}{|\mathcal{A}_t|} \sum_{i \in \mathcal{A}_t} \log \frac{\exp(\text{sim}(P(\mathbf{h}_i), \mathbf{S}_i)/\tau)}{\exp(\text{sim}(P(\mathbf{h}_i), \mathbf{S}_i)/\tau) + \sum_{\delta \in \mathcal{D}_i} \exp(\text{sim}(P(\mathbf{h}_i), \mathbf{S}_{i+\delta})/\tau)}. \quad (7)$$

Gradient balancing. The loss weight λ_{sync} dynamically maintains the synchronization gradient norm at approximately 5% of the flow matching gradient norm, bounded within $[0.02, 0.08]$ after a 5,000-step linear warmup. For audio-only clips, $\mathcal{L}_{\text{sync}}$ is set to zero.

B.3 DYNAMIC PROMPT SCHEDULES

To facilitate mid-stream steering during continuous generation sessions, each 10-s paired training sequence is assigned one of four prompt schedules:

- **Mid-stream switch** (60%): Prompt P_1 conditions the first 5-s segment, switching to prompt P_2 at the 5-s boundary.
- **Instruction withdrawal** (10%): Prompt P_1 conditions the first 5-s segment, followed by generation without text conditioning ($P = \emptyset$) for the second 5 seconds.
- **Instruction arrival** (10%): Generation begins without text conditioning ($P = \emptyset$) for the first 5 seconds, followed by prompt P_1 for the remaining 5 seconds.
- **Sustained hold** (20%): Prompt P_1 is maintained continuously across the full 10-s sequence.

C TRAINING CORPUS AND DATA CURATION

C.1 CORPUS ACCOUNTING AND COMPOSITION

The curated training dataset comprises 993,920 total entries amounting to 1,464.93 hours of audio. Table 12 presents the complete breakdown across all four data pools. An independent held-out validation set of 6,106 entries is isolated from training.

Table 12: Training corpus breakdown by pool and audio duration.

Data Pool	5-s Clips	10-s Paired Views	Total Entries	Audio Duration (h)
Base stereo video-audio	590,510	139,079	590,510	820.15
Mono-dominant video-audio	128,885	48,105	128,885	179.01
Base audio-only	66,859	60,832	127,691	261.84
Recovered audio-only	146,834	—	146,834	203.94
Total Training Inventory	933,088	60,832	993,920	1,464.93

Video-audio 10 s paired views reuse existing 5 s clips and are not counted again in the entry or duration totals. The total row’s 10 s column counts only the 60,832 independent audio-only entries, which are included in both totals.

Source-family breakdown. The base stereo video-audio pool (820.15 h) combines four sources:

- *Open-stereo in-the-wild* (183,271 clips / 254.54 h): VGGSound (Chen et al., 2020) and AudioSet (Gemmeke et al., 2017);
- *Interactive world model* (135,478 clips / 188.16 h): gameplay and simulated scenes;
- *Panoramic ambisonics* (151,298 clips / 210.13 h): Sphere360 (Liu et al., 2025c) (112,502 clips / 156.25 h) and YT-AmbiGen (Kim et al., 2025) (38,796 clips / 53.88 h); and
- *Action-centric and egocentric video* (120,463 clips / 167.31 h): Kinetics-700 (Carreira et al., 2019) (92,474 clips / 128.44 h) and HD-EPIC (Perrett et al., 2025) (27,989 clips / 38.87 h).

The mono-dominant video-audio pool (179.01 h) supplements events from VGGSound (98,357 clips / 136.61 h) and AudioSet (30,528 clips / 42.40 h). These clips retain two-channel audio, with “mono-dominant” referring to duplicated-mono or narrow-stereo signals with limited channel separation. This expansion is screened with Qwen3-Omni and used only during pretraining. It is excluded from spatial fine-tuning. Base audio-only data combines AudioCaps (Kim et al., 2019) and IF-Caps (Tian et al., 2025) (127,691 clips / 261.84 h). The recovered audio-only pool (146,834 clips / 203.94 h) retains weakly-aligned clips lacking direct visual spatial cues but containing clean acoustic events.

C.2 SIGNAL-LEVEL STEREO SCREENING AND AMBISONIC DECODING

The following signal-level screening applies to the base stereo video-audio pool, rather than the separately retained, Qwen3-Omni-screened mono-dominant expansion:

1. **Channel energy and DC centering:** Audio is resampled to 48 kHz and DC-centered per channel. Any clip exhibiting root-mean-square energy $\text{RMS} < 10^{-4}$ on either channel is discarded as silent.
2. **Lateral energy contrast and correlation:** We compute the signed energy contrast $d = (E_R - E_L)/(E_R + E_L)$ and normalized inter-channel cross-correlation $\rho \in [-1, 1]$. Clips with $|\rho| > 0.98$ or stereo width < 0.15 are excluded from the base stereo video-audio pool.

Ambisonic-to-stereo rendering. For Sphere360 and YT-AmbiGen, the soundfield is rotated to align with the primary horizontal source and decoded at $\pm 45^\circ$, matching a 90° perspective crop:

$$L = W + \frac{\sqrt{2}}{2}(X + Y), \quad R = W + \frac{\sqrt{2}}{2}(X - Y). \quad (8)$$

C.3 MULTIMODAL VERIFICATION AND AUDIO-CENTRIC CAPTIONING

Qwen3-Omni (Team, 2025a) scores audio-visual consistency from 1 to 5 and flags non-diegetic music. Video-audio clips below 4, or with such music, are removed. Captions name category, material, direction, reverberation, and envelope, and they omit visual appearance. Video-audio captions see the video and the audio, whereas audio-only captions see the audio only.

C.4 EVALUATION SET ISOLATION AND LEAKAGE AUDIT

Evaluation covers five main held-out test splits: VGGSound (5 s, 10 s) and Interactive Gameplay and Real-World (5 s, 10 s with 4,096 clips each; 30 s with 1,024 clips). We additionally evaluate on Greatest Hits (Owens et al., 2016) physical collisions (244 clips).

We audited the evaluation populations against our training inventory to verify strict data isolation:

- On VGGSound, our evaluation adheres to the official test split, ensuring clip-level isolation against the training corpus.
- On Interactive benchmarks, source-group checks confirmed zero overlap with the training and rollout inventories. The 4,096 clips in the 5-s and 10-s sets derive from 3,111 unique source groups, with at most two non-overlapping windows per source. The 30 s benchmark contains 1,024 distinct held-out sources.

D EVALUATION PROTOCOLS AND METRIC SPECIFICATIONS

D.1 OBJECTIVE METRICS

Metric FD_O computes the Fréchet Audio Distance directly on the mid-channel signal $(L + R)/2$, rather than averaging per-channel distances. Similarly, S-FD_O evaluates the side-channel signal $(L - R)/2$. DeSync is the mean absolute audiovisual synchronization offset, measured in seconds. Lower values indicate better synchronization. Segment-aligned CLAP score details and BiasSkill specifications are presented in Appendix F and Appendix E, respectively. All other objective metrics utilize standard feature extractors as described in Section 5.

D.2 PHYSICAL ONSET EVALUATION PROTOCOL

Greatest Hits (Owens et al., 2016) onset settings follow MMAudio (Cheng et al., 2025): 244 clips, 8 s, 22.05 kHz; threshold $\delta = 0.3$; 50 ms pre/post-peak context; 300 ms matching tolerance.

D.3 HARDWARE BENCHMARK AND STREAMING LATENCY PROTOCOL

Inference latency is measured on a single NVIDIA H100 GPU using batch size 1 and 15 Euler sampling steps per chunk. Steady-state synthesis per 100 ms chunk requires 41.21 ms at p50 and 41.38 ms at p95, achieving a Real-Time Factor (RTF) of 0.41. Over a continuous 10 s streaming session including cold-start initialization, the overall RTF is 0.466. During interactive prompt updates, chunk latency increases slightly to 45.28 ms at p50 and 47.40 ms at p95 due to state cache refresh.

D.4 BASELINE REPRODUCTION AND ACCESS MODE ANALYSIS

We compare WorldSonus against state-of-the-art bidirectional stereo models (AudioX (Tian et al., 2025), ThinkSound (Liu et al., 2025a), PrismAudio (Liu et al., 2025b)) and the streaming monophonic AR baseline V-AURA (Viertola et al., 2025). All baselines are evaluated using official publicly released checkpoints, default samplers, and recommended classifier-free guidance parameters.

Baselines fall into three temporal access modes:

- **Bidirectional access (AudioX, ThinkSound, PrismAudio):** These models require full-clip visual context prior to inference. They process video frames via full-sequence bidirectional attention, rendering them unsuitable for streaming or live-deployment scenarios.
- **Stream mode (V-AURA):** For long-form generation, V-AURA operates with 640 ms strides over a 2.56 s visual window. Within this localized window, frame encoding via TimeS-former (Bertasius et al., 2021) remains non-causal, and DAC (Kumar et al., 2023) neural decoding incorporates future context.
- **Causal mode (WorldSonus):** Visual conditioning is limited to past and current chunk frames, maintaining strict causal alignment as detailed in Section 3.

D.5 ITD-BASED METRICS AND LIMITATIONS

In the production stereo used in our data, channel differences do not reliably reflect physical arrival-time delays. In our diagnostic experiment, duplicating a mono signal across both channels ($L = R$) achieved the lowest ITD-MSE (Chen et al., 2022) among the tested outputs despite having no stereo separation. We therefore omit ITD-MSE from our spatial evaluation. We also exclude FSAD (Sun et al., 2025a) because it uses ITD-related StereoCRW features. Instead, we report S-FD_O and BiasSkill to evaluate stereo energy distribution and left/right channel agreement.

E STEREO BALANCE AGREEMENT (BIASSKILL)

Stereo balance agreement (BiasSkill) measures whether generated and reference audio favor the same left/right channel within 1 s windows. It normalizes the gain over a permutation baseline to account for a model’s fixed channel bias.

Windowing and energy contrast. Audio signals are resampled to 24 kHz and segmented into non-overlapping 1 s windows. Each window is DC-centered per channel. For channel mean-square energies $E_L = \frac{1}{|W|} \sum_{n \in W} x_L[n]^2$ and $E_R = \frac{1}{|W|} \sum_{n \in W} x_R[n]^2$, the left-right energy contrast is:

$$d = \frac{E_R - E_L}{E_R + E_L + 10^{-12}}. \quad (9)$$

A window is designated right-dominant if $d > 0$ and left-dominant if $d < 0$, with $|d|$ reflecting the magnitude of directional separation.

Gating and eligibility criteria. To prevent scoring ambiguous or silent intervals, a ground-truth window is considered eligible only if it satisfies:

$$|d_{\text{GT}}| \geq 0.3, \quad \text{RMS}_{\text{GT}} = \sqrt{\frac{E_L + E_R}{2}} \geq 10^{-4}. \quad (10)$$

For generated audio, we enforce an absolute activity threshold of $\text{RMS}_{\text{gen}} \geq 0.001$ (−60 dBFS). A generated window is marked as correct if and only if:

$$|d_{\text{gen}}| \geq 0.1 \quad \text{and} \quad d_{\text{gen}} \cdot d_{\text{GT}} > 0. \quad (11)$$

Critically, near-mono generations ($|d_{\text{gen}}| < 0.1$) that fail to establish stereo separation are counted as incorrect (retained in the evaluation denominator), preventing models from achieving artificially inflated accuracy by producing centered outputs.

Balanced accuracy. Accuracy is computed over eligible windows with ground-truth side $s \in \{L, R\}$ in clip i , macro-averaged across clips C_s , and balanced between channels:

$$A_s = \frac{1}{|C_s|} \sum_{i \in C_s} \frac{1}{|W_{i,s}|} \sum_{w \in W_{i,s}} \mathbb{1}[\hat{y}_{i,w} = s], \quad A_{\text{bal}} = \frac{1}{2}(A_L + A_R). \quad (12)$$

Table 13 summarizes the eligible window counts and retention rates across test sets.

Table 13: Eligible 1 s window counts and retention rates for BiasSkill evaluation.

Set	Clips	Cand. Win.	Elig. Clips	Elig. Win. (L / R)	Keep (%)
VGGSound 10 s	4,096	40,960	1,589	3,314 / 3,356	16.3
Interactive 10 s	4,096	40,960	2,275	6,759 / 3,316	24.6
Interactive 30 s	1,024	30,720	679	2,870 / 3,200	19.8
Total	9,216	112,640	—	12,943 / 9,872	20.3

Permutation null and normalized BiasSkill. Generative models often exhibit fixed channel biases stemming from training data or architecture (e.g., ThinkSound consistently generates right-heavy energy, whereas PrismAudio is left-heavy). To eliminate these baseline preferences, we draw 10,000 whole-clip derangements shared across all models, pairing generated audio clips with mismatched ground-truth videos while maintaining internal window alignment. The permutation null score μ_{null} represents the expected balanced accuracy under these shuffled pairings. BiasSkill is defined as the normalized gain over this model-specific null:

$$\text{BiasSkill} = \frac{A_{\text{bal}} - \mu_{\text{null}}}{1 - \mu_{\text{null}}} \times 100\%. \quad (13)$$

A BiasSkill score of 0% indicates performance no better than random chance under the model’s fixed channel bias, while 100% represents perfect left/right channel agreement on eligible windows. The one-sided Monte Carlo p -value is given by:

$$p = \frac{1 + \sum_{b=1}^B \mathbb{1}[A_{\text{bal}}^{(b)} \geq A_{\text{bal}}]}{B + 1}, \quad (14)$$

with $B = 10,000$. Table 14 reports the detailed stereo balance agreement results.

F INTERACTIVE TEXT CONTROL AND PROMPT-SWITCH ANALYSIS

F.1 CLAP IMPLEMENTATION AND SEGMENT ALIGNMENT

Semantic audio-text correspondence is evaluated using LAION-CLAP (Wu et al., 2023). Audio tracks are converted to mono at a 48 kHz sampling rate. Each waveform is divided into consecutive, non-overlapping 5 s segments and evaluated against its corresponding audio caption. Embeddings are extracted in FP16 precision and normalized in FP32. We report raw cosine similarities, macro-averaged within individual clips and across the test evaluation set.

Table 14: Detailed stereo balance agreement results under the -60 dBFS generation-RMS gate. L/R Rec. denotes per-side recall; Bal. Acc. is A_{bal} ; RMS Cov. is the coverage of $\text{RMS}_{\text{gen}} \geq 0.001$; Act. Cov. requires both RMS and $|d_{\text{gen}}| \geq 0.1$; $\Delta = A_{\text{bal}} - \mu_{\text{null}}$; BiasSkill represents the normalized effect size (%); and p indicates the one-sided permutation test value.

Set	Model	L Rec.	R Rec.	Bal. Acc.	RMS Cov.	Act. Cov.	Null μ_{null}	Δ	BiasSkill (%)	p
VGG 10 s	AudioX	0.273	0.186	0.229	0.990	0.430	0.199	+0.030	3.79	0.0002
	ThinkSound	0.181	0.386	0.283	0.956	0.554	0.260	+0.024	3.22	0.0035
	PrismAudio	0.277	0.093	0.185	0.946	0.364	0.176	+0.009	1.10	0.1052
	WorldSonus (Ours)	0.256	0.211	0.233	0.976	0.439	0.198	+0.035	4.42	0.0001
Int. 10 s	AudioX	0.256	0.237	0.246	0.967	0.460	0.222	+0.024	3.08	0.0001
	ThinkSound	0.159	0.395	0.277	0.983	0.594	0.290	-0.013	-1.78	0.9568
	PrismAudio	0.387	0.036	0.211	0.995	0.447	0.212	-0.001	-0.06	0.5390
	WorldSonus (Ours)	0.317	0.273	0.295	0.995	0.559	0.250	+0.045	6.03	0.0001
Int. 30 s	AudioX	0.247	0.225	0.236	0.999	0.468	0.236	0.000	-0.01	0.4974
	ThinkSound	0.101	0.455	0.278	0.997	0.559	0.273	+0.005	0.63	0.3088
	PrismAudio	0.493	0.046	0.270	0.997	0.519	0.238	+0.032	4.19	0.0004
	WorldSonus (Ours)	0.447	0.397	0.422	0.995	0.716	0.313	+0.109	15.90	0.0001

F.2 PAIRED COUNTERFACTUAL INTERVENTION PROTOCOL

To rigorously verify that dynamic acoustic changes during streaming stem from active prompt steering rather than underlying visual transitions in the video, we construct a paired counterfactual intervention across all 4,096 clips in VGGSound 10 s and Interactive 10 s.

For each video clip with consecutive reference prompts (P_1, P_2) , we synthesize two paired generations sharing identical model parameters (150 k EMA checkpoint), visual feature inputs, random seeds, Euler solver settings (15 steps), and classifier-free guidance scales:

- **Switch Arm:** Conditioned on prompt P_1 for 0–5 s, dynamically switching to prompt P_2 at the 5 s boundary without clearing model state cache.
- **Hold Arm:** Conditioned continuously on prompt P_1 throughout the full 0–10 s duration.

Let A_1 and A_2 denote the generated 5 s audio segments, and let $s_{ij} = \cos(e_A(A_i), e_T(P_j))$ represent the pairwise cosine similarity between audio segment A_i and text prompt P_j . Segment alignment S , matching advantage M , and dual relative match indicator R are formulated as:

$$S = \frac{1}{2}(s_{11} + s_{22}), \quad (15)$$

$$M = \frac{1}{2}[(s_{11} - s_{12}) + (s_{22} - s_{21})], \quad (16)$$

$$R = \mathbb{K}[s_{11} > s_{12} \wedge s_{22} > s_{21}]. \quad (17)$$

The second segment’s preference for the updated prompt is given by $\Delta = s_{22} - s_{21}$. The net paired intervention gain G_i for clip i is defined as:

$$G_i = \Delta_i^{\text{switch}} - \Delta_i^{\text{hold}}. \quad (18)$$

Table 15 summarizes the full paired intervention experimental results.

F.3 CONFIDENCE INTERVALS AND PROMPT STRATIFICATION

Confidence intervals for the mean intervention gain G are derived via 20,000 paired bootstrap resamples of clips within each benchmark dataset (random seed 5031). We report the 2.5th and 97.5th percentiles as $[l, u]$. The compact $\pm$ margin corresponds to half of this interval width.

To evaluate how semantic contrast influences prompt control responsiveness, clips are ranked by text-to-text cosine similarity $\cos(e_T(P_1), e_T(P_2))$ and partitioned into four equal quartiles of 1,024 clips each (Table 16). Quartile Q1 contains the most semantically distinct prompt pairs, while Q4 contains the most similar. As shown, G remains positive across all strata and scales directly with the degree of prompt divergence.

Table 15: Full paired-control intervention results (4,096 clips per set). The s_{ij} entries are raw CLAP cosine similarities. S is the mean of the aligned similarities s_{11} and s_{22} , M is the mean matching advantage, and R is reported as a percentage (%). Hold is evaluated against the same reference prompt pair (P_1, P_2) as Switch.

Set	Arm	s_{11}	s_{12}	s_{21}	s_{22}	S	M	R (%)
VGG 10 s	Switch	0.3823	0.3312	0.3293	0.3530	0.3677	0.0374	25.68
	Hold	0.3823	0.3312	0.3543	0.3270	0.3546	0.0119	14.43
Int. 10 s	Switch	0.3560	0.3062	0.3062	0.3284	0.3422	0.0360	23.44
	Hold	0.3560	0.3062	0.3316	0.3013	0.3286	0.0097	12.82

Table 16: Stratification of paired prompt gain G across text-similarity quartiles (1,024 clips per quartile). Brackets denote 95% clip-level bootstrap percentile intervals.

Evaluation Set	Quartile	Text Cosine Range	Intervention Gain G
VGGSound 10 s	Q1 (Most Distinct)	$[-0.244, 0.525]$	0.1393 $[0.1287, 0.1499]$
	Q2	$[0.525, 0.727]$	0.0441 $[0.0388, 0.0494]$
	Q3	$[0.727, 0.842]$	0.0157 $[0.0127, 0.0187]$
	Q4 (Most Similar)	$[0.842, 0.990]$	0.0047 $[0.0030, 0.0064]$
Interactive 10 s	Q1 (Most Distinct)	$[-0.287, 0.490]$	0.1277 $[0.1187, 0.1367]$
	Q2	$[0.491, 0.688]$	0.0514 $[0.0459, 0.0569]$
	Q3	$[0.688, 0.820]$	0.0247 $[0.0210, 0.0284]$
	Q4 (Most Similar)	$[0.820, 0.990]$	0.0063 $[0.0046, 0.0080]$

G SUBJECTIVE USER STUDY PROTOCOL

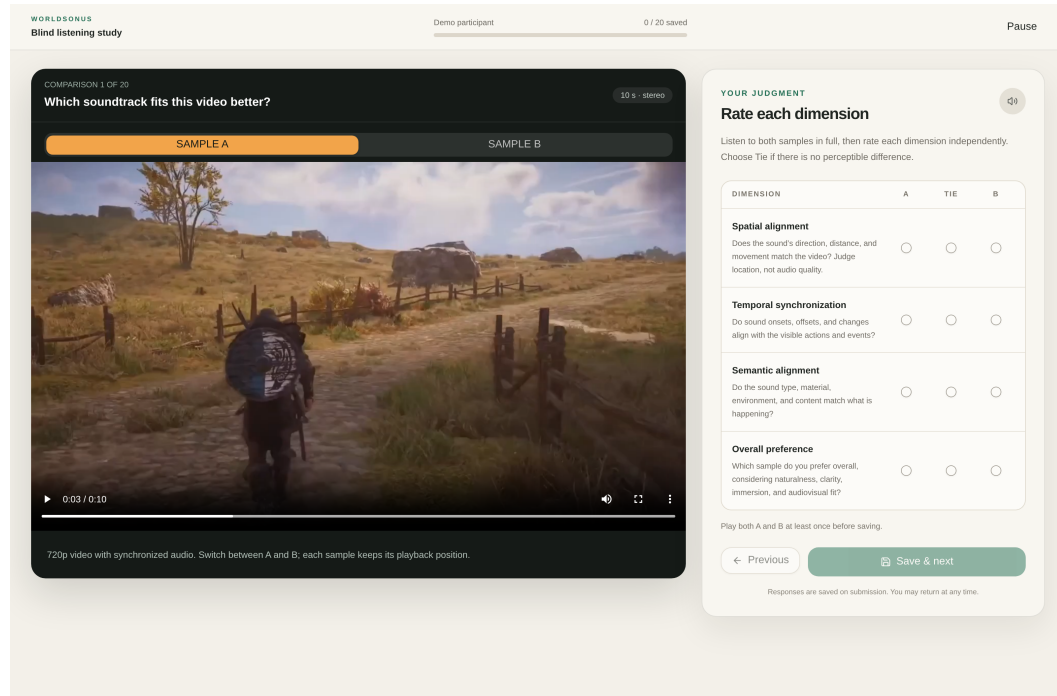
The user study evaluates 40 clips from Interactive 10 s, comprising 20 gameplay and 20 real-world scenes. Participants evaluate randomized, anonymized A/B pairs (with a tie option) across four dimensions: spatial alignment, temporal synchronization, semantic consistency, and overall preference. The five evaluated audio sources comprise Ground Truth, WorldSonus, AudioX, ThinkSound, and PrismAudio (V-AURA is excluded due to its monophonic output). Twenty independent evaluators produced a total of 400 pairwise evaluations, yielding 40 ratings per paired comparison.

Figure 4 shows the evaluation interface for gameplay and real-world clips, with paired samples labeled A and B and a separate A/Tie/B choice for each criterion.

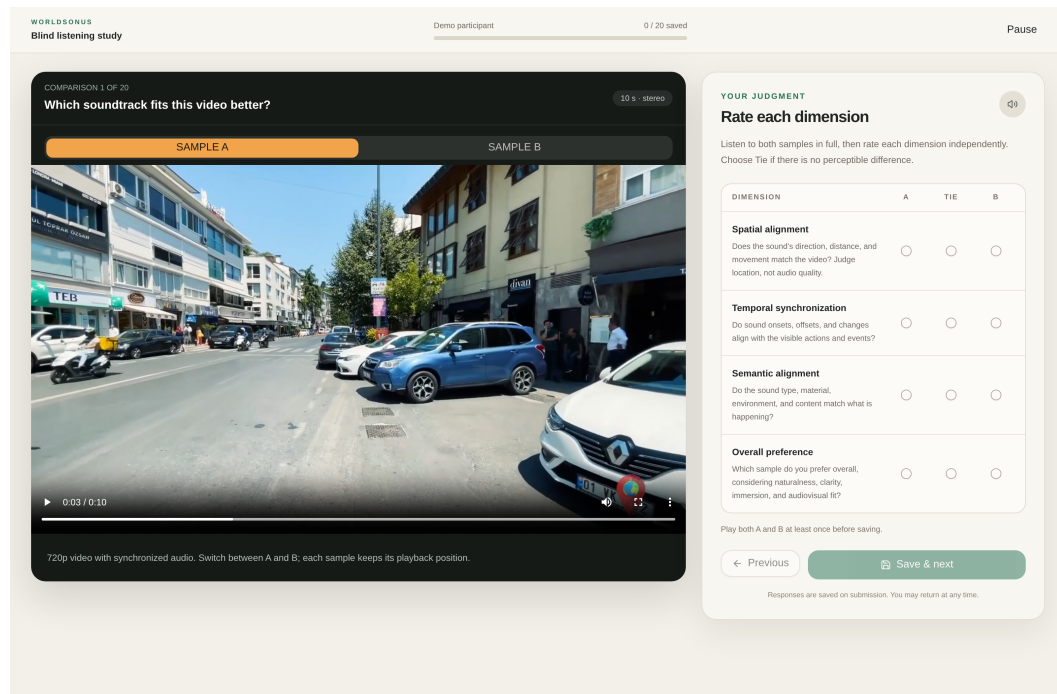
The preference score $P_{ij}^{(d)}$ of method i over method j on criterion d is calculated as:

$$P_{ij}^{(d)} = 100 \times \frac{W_{ij}^{(d)} + \frac{1}{2}T_{ij}^{(d)}}{W_{ij}^{(d)} + T_{ij}^{(d)} + L_{ij}^{(d)}}, \quad (19)$$

where $W_{ij}^{(d)}$, $T_{ij}^{(d)}$, and $L_{ij}^{(d)}$ denote total wins, ties, and losses, respectively. Ties contribute half a vote (0.5), ensuring symmetric scoring $P_{ij}^{(d)} + P_{ji}^{(d)} = 100\%$. All evaluation matrices are reported on a normalized 0–100% scale.



(a) Gameplay example



(b) Real-world example

Figure 4: User-study interface for gameplay and real-world clips. Participants compare samples A and B and select A, Tie, or B for spatial alignment, temporal synchronization, semantic alignment, and overall preference.